

Додаток 2. Результати тестування

Basic

examples/basic.signal

```
1 PROGRAM HELLO;  
2 CONST  
3   X = '42';  
4 BEGIN  
5 END.
```

Output

Tokens

Code	Row	Col	Type	Value
256	1	1	Keyword	PROGRAM
1000	1	9	Identifier	HELLO
59	1	14	Delimiter	;
257	2	1	Keyword	CONST
1001	3	3	Identifier	X
61	3	5	Delimiter	=
39	3	7	Delimiter	'
500	3	8	Literal	42
39	3	10	Delimiter	'
59	3	11	Delimiter	;
258	4	1	Keyword	BEGIN
259	5	1	Keyword	END
46	5	4	Delimiter	.

Identifiers

Code	Value
1000	HELLO
1001	X

Literals

Code	Value
500	42

Comments

examples/comments.signal

```
1 (* A lot of comments *)
2 PROGRAM COMMENTS;
3 CONST
4   (* Single-line comment *)
5   (*
6     Multi-line comment
7   *)
8   (**) (* *)
9   (* Multiple *) (* comments *) (* on *) (* the *) (* same *) (* line *)
10  (****) ((*())*) (*;.:*) (* Comments with nested * and () *)
11  (* Invalid symbols are allowed! %#^@& *)
12 BEGIN (* on the same like as identifier *)
13 END (* between tokens *) .
```

Output

Tokens					
Code	Row	Col	Type	Value	
256	2	1	Keyword	PROGRAM	
1000	2	9	Identifier	COMMENTS	
59	2	17	Delimiter	;	
257	3	1	Keyword	CONST	
258	12	1	Keyword	BEGIN	
259	13	1	Keyword	END	
46	13	26	Delimiter	.	

Identifiers	
Code	Value
1000	COMMENTS

Literals	
Code	Value

Constants

examples/constants.signal

```
1 (* All ways to declare a constant within a grammar *)
2 PROGRAM CONSTANTS;
3 CONST
4   (* Single-line comment *)
5   A = '';
6   A = '1';
7   A = '2,3';
8   B = ',4';
9   B = '5$EXP(6)';
10  C = '$EXP(7)';
11 BEGIN
12   (*
13    Multi-line comment
14   *)
15 END.
```

Output

Tokens					
Code	Row	Col	Type	Value	
256	2	1	Keyword	PROGRAM	
1000	2	9	Identifier	CONSTANTS	
59	2	18	Delimiter	;	
257	3	1	Keyword	CONST	
1001	5	3	Identifier	A	
61	5	5	Delimiter	=	
39	5	7	Delimiter	'	
39	5	8	Delimiter	'	
59	5	9	Delimiter	;	
1001	6	3	Identifier	A	
61	6	5	Delimiter	=	
39	6	7	Delimiter	'	
500	6	8	Literal	1	
39	6	9	Delimiter	'	
59	6	10	Delimiter	;	
1001	7	3	Identifier	A	
61	7	5	Delimiter	=	
39	7	7	Delimiter	'	
501	7	8	Literal	2	
44	7	9	Delimiter	,	
502	7	10	Literal	3	
39	7	11	Delimiter	'	
59	7	12	Delimiter	;	
1002	8	3	Identifier	B	
61	8	5	Delimiter	=	
39	8	7	Delimiter	'	
44	8	8	Delimiter	,	
503	8	9	Literal	4	
39	8	10	Delimiter	'	
59	8	11	Delimiter	;	
1002	9	3	Identifier	B	
61	9	5	Delimiter	=	
39	9	7	Delimiter	'	
504	9	8	Literal	5	
36	9	9	Delimiter	\$	
260	9	10	Keyword	EXP	
40	9	13	Delimiter	(	
505	9	14	Literal	6	
41	9	15	Delimiter	)	
39	9	16	Delimiter	'	

Edge Cases

examples/edge-cases.signal

```
1 (* Edge cases within the grammar *)
2 PROGRAM EDGES;
3 CONST
4   AB12CD = '999,000';
5   Z = '100$EXP(200)';
6   LONGNAME123 = '0';
7   A = '0,0';
8 BEGIN
9 END.
```

Output

Tokens					
Code	Row	Col	Type	Value	
256	2	1	Keyword	PROGRAM	
1000	2	9	Identifier	EDGES	
59	2	14	Delimiter	;	
257	3	1	Keyword	CONST	
1001	4	3	Identifier	AB12CD	
61	4	10	Delimiter	=	
39	4	12	Delimiter	'	
500	4	13	Literal	999	
44	4	16	Delimiter	,	
501	4	17	Literal	000	
39	4	20	Delimiter	'	
59	4	21	Delimiter	;	
1002	5	3	Identifier	Z	
61	5	5	Delimiter	=	
39	5	7	Delimiter	'	
502	5	8	Literal	100	
36	5	11	Delimiter	\$	
260	5	12	Keyword	EXP	
40	5	15	Delimiter	(	
503	5	16	Literal	200	
41	5	19	Delimiter	)	
39	5	20	Delimiter	'	
59	5	21	Delimiter	;	
1003	6	3	Identifier	LONGNAME123	
61	6	15	Delimiter	=	
39	6	17	Delimiter	'	
504	6	18	Literal	0	
39	6	19	Delimiter	'	
59	6	20	Delimiter	;	
1004	7	3	Identifier	A	
61	7	5	Delimiter	=	
39	7	7	Delimiter	'	
504	7	8	Literal	0	
44	7	9	Delimiter	,	
504	7	10	Literal	0	
39	7	11	Delimiter	'	
59	7	12	Delimiter	;	
258	8	1	Keyword	BEGIN	
259	9	1	Keyword	END	
46	9	4	Delimiter	.	

Identifiers	
Code	Value
1000	EDGES

	1001		AB12CD	
	1002		Z	
	1003		LONGNAME123	
	1004		A	
+-----+-----+-----+				

Literals

+-----+-----+-----+				
	Code		Value	
+-----+-----+-----+				
	500		999	
	501		000	
	502		100	
	503		200	
	504		0	
+-----+-----+-----+				

Empty

examples/empty.signal

```
1 (* Empty program *)
2 PROGRAM EMPTY;
3 BEGIN
4 END.
```

Output

Tokens					
+	-----	+	-----	+	-----
	Code		Row		Col
	Type		Value		
+	-----	+	-----	+	-----
	256		2		1
	Keyword		PROGRAM		
	1000		2		9
	Identifier		EMPTY		
	59		2		14
	Delimiter		;		
	258		3		1
	Keyword		BEGIN		
	259		4		1
	Keyword		END		
	46		4		4
	Delimiter		.		
+	-----	+	-----	+	-----
Identifiers					
+	-----	+	-----	+	-----
	Code		Value		
+	-----	+	-----	+	-----
	1000		EMPTY		
+	-----	+	-----	+	-----
Literals					
+	-----	+	-----	+	-----
	Code		Value		
+	-----	+	-----	+	-----
+	-----	+	-----	+	-----

Error Comment Not Closed

examples/error-comment-not-closed.signal

```
1 (* Check comment not closed error *)
2 PROGRAM NOCLOSE;
3 BEGIN
4 END.
5 (* No closing bracket!
```

Output

Tokens					
+	-----	+	-----	+	-----
	Code		Row		Col
	Type		Value		
+	-----	+	-----	+	-----
	256		2		1
	Keyword		PROGRAM		
	1000		2		9
	Identifier		NOCLOSE		
	59		2		16
	Delimiter		;		
	258		3		1
	Keyword		BEGIN		
	259		4		1
	Keyword		END		
	46		4		4
	Delimiter		.		
+	-----	+	-----	+	-----
Tokenizer Error [5:1]: Comment not closed					
Identifiers					
+	-----	+	-----	+	-----
	Code		Value		
+	-----	+	-----	+	-----
	1000		NOCLOSE		
+	-----	+	-----	+	-----
Literals					
+	-----	+	-----	+	-----
	Code		Value		
+	-----	+	-----	+	-----
+	-----	+	-----	+	-----

Error Invalid Character

examples/error-invalid-character.signal

```
1 (* Encountered invalid character(s) *)
2 PROGRAM INVCHARS;
3 CONST
4   A = '1,%'; (* 1 error: % is invalid *)
5   b#& = '30$EXP(40)'; (* 3 errors: b, #, & are invalid *)
6 BEGIN (* 1 error: grammar is case-sensitive! *)
7 END.
```

Output

Tokens					
Code	Row	Col	Type	Value	
256	2	1	Keyword	PROGRAM	
1000	2	9	Identifier	INVCHARS	
59	2	17	Delimiter	;	
257	3	1	Keyword	CONST	
1001	4	3	Identifier	A	
61	4	5	Delimiter	=	
39	4	7	Delimiter	'	
500	4	8	Literal	1	
44	4	9	Delimiter	,	
39	4	11	Delimiter	'	
59	4	12	Delimiter	;	
61	5	7	Delimiter	=	
39	5	9	Delimiter	'	
501	5	10	Literal	30	
36	5	12	Delimiter	\$	
260	5	13	Keyword	EXP	
40	5	16	Delimiter	(	
502	5	17	Literal	40	
41	5	19	Delimiter	)	
39	5	20	Delimiter	'	
59	5	21	Delimiter	;	
1002	6	1	Identifier	BEGI	
259	7	1	Keyword	END	
46	7	4	Delimiter	.	
Tokenizer Error [4:10]: Invalid character '%'					
Tokenizer Error [5:3]: Invalid character 'b'					
Tokenizer Error [5:4]: Invalid character '#'					
Tokenizer Error [5:5]: Invalid character '&'					
Tokenizer Error [6:5]: Invalid character 'n'					
Identifiers					
Code	Value				
1000	INVCHARS				
1001	A				
1002	BEGI				
Literals					
Code	Value				
500	1				
501	30				
502	40				

Error Mixed

examples/error-mixed.signal

```
1 (* Multiple error types in one file *)
2 PROGRAM MIXED;
3 CONST
4   A = '1';
5   bad = '2'; (* lowercase: 3 errors *)
6   C = '3~4'; (* invalid ~: 1 error *)
7 BEGIN
8 END.
9 (* unclosed comment
```

Output

Tokens					
Code	Row	Col	Type	Value	
256	2	1	Keyword	PROGRAM	
1000	2	9	Identifier	MIXED	
59	2	14	Delimiter	;	
257	3	1	Keyword	CONST	
1001	4	3	Identifier	A	
61	4	5	Delimiter	=	
39	4	7	Delimiter	'	
500	4	8	Literal	1	
39	4	9	Delimiter	'	
59	4	10	Delimiter	;	
61	5	7	Delimiter	=	
39	5	9	Delimiter	'	
501	5	10	Literal	2	
39	5	11	Delimiter	'	
59	5	12	Delimiter	;	
1002	6	3	Identifier	C	
61	6	5	Delimiter	=	
39	6	7	Delimiter	'	
502	6	8	Literal	3	
503	6	10	Literal	4	
39	6	11	Delimiter	'	
59	6	12	Delimiter	;	
258	7	1	Keyword	BEGIN	
259	8	1	Keyword	END	
46	8	4	Delimiter	.	
Tokenizer Error [5:3]: Invalid character 'b'					
Tokenizer Error [5:4]: Invalid character 'a'					
Tokenizer Error [5:5]: Invalid character 'd'					
Tokenizer Error [6:9]: Invalid character '~'					
Tokenizer Error [9:1]: Comment not closed					
Identifiers					
Code	Value				
1000	MIXED				
1001	A				
1002	C				
Literals					
Code	Value				
500	1				
501	2				

	502		3	
	503		4	
+-----+-----+				